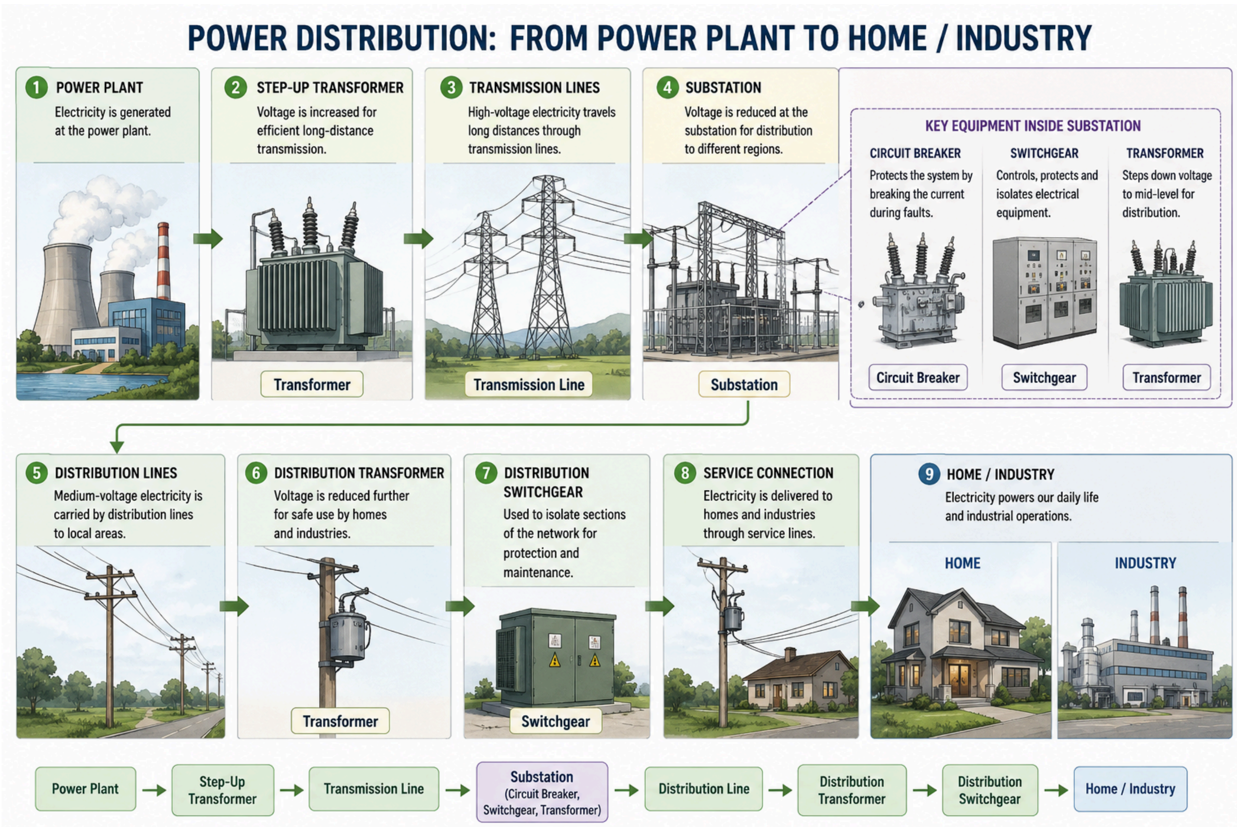
Electricity Distribution

Domain Knowledge

High Level Picture depicting the distribution of power



Components Involved during transmission

Data (asset type) given in our dataset:

1. Circuit Breaker
2. Substation
3. Switchgear
4. Transformer
5. Transmission line

Transformer

Purpose

Transformers change voltage levels.

Failure Causes

1. Insulation degradation
2. Oil contamination
3. Overheating
4. Moisture ingress
5. Overloading

Transmission Line

Purpose

Carry electricity over long distances.

Failure Causes

1. Storms
2. Lightning
3. Tree contact
4. Conductor damage
5. Tower collapse

Substation

Purpose

Substation acts like a traffic junction for electricity

Functions:

Voltage transformation
Power routing
Fault isolation
Grid monitoring

Failure Causes

- Transformer failures (causes given under Transformer section)

- Overloading - Demand exceeds the designed capacity of the substation
- Circuit Breaker failure (causes given under Circuit Breaker)
- Switch gear failure (causes given under Circuit Breaker)
- Weather events
- Aging Infra
- Poor Maintenance

Circuit Breaker

Purpose

Protect the grid - isolate the problem

Fault Causes

1. Mechanical wear
2. contact degradation
3. control failures

Switchgear

Purpose

Control and isolate electrical equipment. You can think of switchgear as the "traffic controller" of the grid.

Failure Causes

1. Arc flash
2. Insulation failure
3. Aging components